

Department of Computer Science and Telecommunication Engineering

Noakhali Science and Technology University

Year-4, Term-II; Final Examination-2026

Session: 2020-2021

~~ECO~~ ^{HUM} 4201: Principles of Economics

Time: 3 hours

Credit: 2

Full Marks: 70

Instructions

1. Answer any seven questions.

2. Draw diagrams where necessary

1. (a) Define economics and explain its major definitions: wealth-based, welfare-based, scarcity-based, and growth-and-development-based definitions. 4
- (b) Discuss whether software, data, cloud storage, online courses, and AI tools can be considered economic goods. 3
- (c) Distinguish between money and near money. 3
2. (a) State the law of demand and write a complete demand function. 2
- (b) Explain, using a suitable diagram, how equilibrium is determined in the market for online programming courses. 4
- (c) Using separate diagrams, show how equilibrium changes in each of the following cases: 4
 - i) An increase in freelancing opportunities affects the market for mobile data.
 - ii) An increase in the cost of online learning affects the market for software developers.
3. Using your knowledge of market structures, answer the following:
 - i) How can network effects create monopoly power in messaging apps? 2
 - ii) Why does enterprise software pricing differ from student pricing? 2
 - iii) How does non-price competition work in smartphone markets? 2
 - iv) Why can a highly rated freelancer charge more than a beginner? 2
 - v) How does monopolistic competition apply to YouTube channels? 2
4. (a) List the characteristics of a good tax system. 3
- (b) Discuss the role of e-filing, digital payments, e-invoicing, data analytics, artificial intelligence, blockchain, and cybersecurity in tax administration. 7

Please turn over

5. Consider the following hypothetical data of national income accounting:

A. Items	B. Quantity		C. Price/value ('000)		D. Additional Data for 2025	
	2024	2025	2024	2025	Items	Value
Firm equipment	20	25	10	12	Income abroad	25000
Drugs	100	120	0.10	0.12	Income to foreigners	20000
Salaries from govt.	200	210	1.00	1.20	Depreciation	15000
Restaurant meals	400	350	0.05	0.06	Statistical discrepancy	10000
Home-cooked meals	500	550	0.05	0.06	Corporate taxes	8000
University tuition fees	300	320	0.20	0.22	Retained earnings	7000
Imported mobile	200	180	0.50	0.55	Transfers	40000
Second-hand mobile	50	60	0.30	0.35	Personal taxes	30000
New house	10	8	50	60		
Road construction	---	---	50	70		
Machinery imports	---	---	40	60		
Pension	---	---	30	40		
Inventory	---	---	20	30		
Stocks	---	---	15	18		
Grain (domestic)	1000	1200	0.01	0.012		
RMG exports to USA	---	---	80	100		
Homestead vegetables	---	---	10	15		
Airplane from Boeing	2	3	100	100		

- Classify each item of column A into C, I, G, NX or Excluded 2
 - Compute nominal GDP (2024 & 2025) 3
 - Determine real GDP (Base year = 2024) [Note: Assume constant price for the mentioned total expenditure (value) accounts. 2
 - Calculate GDP deflator & inflation rate 1
 - Using additional data (column D), estimate: GNP, NNP, NI, PI, and DPI. 2
6. (a) Suppose the base year basket consists of: Rice = 100 units (Price = Tk. 10) and Cars = 10 units (Price = Tk. 100). In the current year: Rice price = Tk. 12 and Car price = Tk. 150. 2
- Calculate the CPI for the current year (base year = 100). 1
 - Calculate the inflation rate. 1
- (b) Now assume current year quantities change due to substitution: Rice = 120 units, and Cars = 8 units. 2
- Calculate the GDP deflator. 1
 - Explain why CPI and GDP deflator give different inflation rates in this case. 1
 - Identify which measure better reflects consumer welfare and justify your answer. 1
- (c) Discuss how the following three problems affect CPI as a measure of cost of living: (i) Substitution bias, (ii) Introduction of new goods, (iii) Unmeasured quality change. Then critically evaluate: "CPI systematically overstates inflation and may lead to incorrect policy decisions." Support your answer using numerical intuition from earlier parts. 3

Please turn over

7. Tech Support Ltd. employs entry-level programmers. The equilibrium wage is Tk. 30,000 per month. The government sets a minimum wage of Tk. 36,000. At this wage, 180 programmers seek jobs, but the firm hires only 140.

- i) Explain how a minimum wage above the equilibrium wage creates unemployment. 3
- ii) Calculate the number of unemployed programmers created by the minimum wage. Illustrate your answer using labor demand and labor supply concepts. 3
- iii) Evaluate whether minimum-wage laws should be increased for the IT sector. Present two arguments in favor and two against. 4

8. An AI software company adopts a new programming language. Some programmers leave their jobs voluntarily to find better opportunities, while others lose jobs because their skills no longer match market demand.

- (a) Distinguish between frictional and structural unemployment. 3
- (b) Explain why the unemployment rate never permanently falls to zero even in a healthy economy. 3
- (c) Identify the type of unemployment experienced by each group in the stimulus and recommend one policy to reduce each type. 4

9. (a) A cloud-computing company observes that Bangladesh Bank purchases government securities, increasing the money supply.

- i) Explain the roles of money supply and money demand in determining the price level. 2
 - ii) Draw and explain the money-market equilibrium diagram, indicating the money supply and money demand curves. 2
 - iii) Illustrate, step by step, how an increase in the money supply changes equilibrium and affects the value of money and the price level. 3
- (b) What are the costs of inflation? Which of these costs do you think are most important for the Bangladesh economy. 3